



Walchand Institute of Technology, Solapur
(An Autonomous Institute)

Affiliated to
Punyashlok Ahilyadevi Holkar Solapur University,
Solapur

Choice Based Credit System (CBCS)

Structure and Syllabus
for

F.Y.B.Tech. Group C

(Computer Science and Engineering, Information Technology,
Artificial Intelligence & Machine Learning)

W.E.F. 2026-27



HEAD

General Engg. Department
Walchand Institute of
Technology, Solapur


Dean(Academics)



General Engineering Department

Vision

- To produce young, globally competent Graduates / Post-Graduates / Doctoral Engineers with an aptitude for leadership and research, to face the challenges for modernization and globalization courageously who will be instrumental for overall growth of the society.

Mission

- To impart quality Technical Education in accordance with the needs of the society through various academic programs.
- To foster learning process & to provide proper ambience for motivating students for creating awareness to excel in the ever expanding field of science and technology.
- To enhance career opportunities for students through exposure to industries & research institutions.
- To strive for excellence by encouraging independent critical thinking, creativity & discipline.
- To create awareness for engineering ethics & human values for instilling moral, social values & loyalty & to appreciate the rights of others & respect towards society & its heritage.
- To help the students to implement their acquired Engineering knowledge for society & community development, thus, enhancing a strong sense of social responsibility & accountability.
- To reach to the community through various outreach programs to include the scientific technological spirit among all.
- To promote & provide a framework to meet campus sustainability goals & mitigate climate change.
- To help in nation building through a pool of dedicated, disciplined, intellectual & integrated manpower.

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General Engineering Department

Legends Used

L	Lecture Hours / week
T	Tutorial Hours / week
P	Practical Hours / week
FA	Formative Assessment
SA	Summative Assessment
ICA	Internal Continuous Assessment
POE	Practical and Oral Exam
OE	Oral Exam
MOOC	Massive Open Online Course
HSS	Humanity and Social Science
NPTEL	National Programme on Technology Enhanced Learning
F.Y.	First Year
S.Y.	Second Year
T.Y.	Third Year
B. Tech.	Bachelor of Technology

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QSC
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General Engineering Department

Course Code Format

G	E	1	C	C	1	L
Program Code		Semester No./ Year 1/2/3/...8	Course Type		Course Serial No. 1-9	L-Lab. session

Program Code	
GE	General Engineering
Course Type	
BS	Basic Science
ES	Engineering Science
HU	Humanities & Social Science
CC	Co-Curricular Course
PC	Professional Core Course
MC	Mandatory Course
IK	Indian Knowledge System
AE	Ability Enhancement Course
VS	Vocational and Skill Enhancement Course

Sample Course Code	
GEIBS1	Engineering Physics

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Walchand Institute of Technology, Solapur E Y B Tech. (Group C) Syllabus (Scheme 2020-21)


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Page 2



General Engineering Department

First Year B. Tech. Semester I (Group C) Computer Science and Engineering, Information Technology, Artificial Intelligence & Machine Learning

• Theory Courses:

Course Code	Name of Course	Engagement Hours			Credits	SA	FA		Total
		L	T	P		ESE	ISE	ICA	
GE1BS2	Engineering Chemistry	2	-	-	2	60	40	-	100
GE1BS3	Engg. Mathematics-I	2	1	-	3	60	40	25	125
EC1ES1	Basic Electrical & Electronics Engineering	2	-	-	2	60	40	-	100
GE1AE1	Professional & Communication Skills	2	1	-	3	60	40	25	125
GE1IK1	Indian Science & Technology	2	-	-	2	-	25	25	50
GE1CC1N*	Co-curricular Activity	-	-	-	1	-	-	25	25
Total		10	2	-	13	240	185	100	525

• Laboratory Courses:

Course Code	Name of Course	Engagement Hours			Credits	SA	FA		Total
		L	T	P		ESE	ISE	ICA	
GE1BS2L	Engineering Chemistry	-	-	2	1	-	-	25	25
EC1ES1L	Basic Electrical & Electronics Engineering	-	-	2	1	-	-	25	25
CS1PC1L	Programming in C	2	-	2	3	50#	25	25	100
GE1CC2L	Creativity & Design Thinking	1	-	2	2	-	-	50	50
Total		3	-	8	7	50	25	125	200
Grand Total		13	2	8	20	290	210	225	725

Note:

- # Indicates the subject 'C Programming' shall have a Practical and Oral Examination at the end of the semester.
- N* :The Name of Courses and Course Codes for Co-curricular Activity will be as below

Course Code	Name of Course
GE1CC11	Yoga
GE1CC12	Health and Wellness
GE1CC13	Sports

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Walchand Institute of Technology, Solapur: FYB Tech. (Group C) Syllabus (Scheme 2020-27)



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First Year B. Tech. Semester II (Group C)

• Theory Courses:

Course Code	Name of Course	Engagement Hours			Credits	SA	FA		Total
		L	T	P		ESE	ISE	ICA	
GE2BS1	Engineering Physics	2	-	-	2	60	40	-	100
GE2BS4	Engg. Mathematics-II	2	1	-	3	60	40	25	125
CE2ES2	Basic Civil & Mechanical Engineering	3	-	-	3	60	40	-	100
MA2ES3	Engg. Graphics & CAD	2	-	-	2	-	50	-	50
GE2VS1	English for Employability Skills	2	1	-	3	60	40	25	125
Total		11	2	-	13	240	210	50	500

• Laboratory Courses:

Course Code	Name of Course	Engagement Hours			Credits	SA	FA		Total
		L	T	P		ESE	ISE	ICA	
GE2BS1L	Engineering Physics	-	-	2	1	-	-	25	25
CE2ES2L	Basic Civil & Mechanical Engineering	-	-	2	1	-	-	25	25
MA2ES3L	Engg. Graphics & CAD	-	-	2	1	-	-	50	50
CS2PC2L	Python Programming	2	-	2	3	50#	25	25	100
GE2VS2L	Workshop Practice	1	-	2	2	-	-	50	50
Total		3	-	10	8	50	25	175	250
Grand Total		14	2	10	21	290	235	225	750

• Mandatory Courses:

Course Code	Name of Course	Engagement Hours			Credits	SA	FA		Total
		L	T	P		ESE	ISE	ICA	
GE2MC1	Democracy, Elections and Good Governance	-	-	-	-	50	-	-	50
GE2MC2	Disaster Management	-	-	-	-	-	-	50	50

Note:

- # Indicates the subject 'Python Programming' shall have a Practical and Oral Examination at the end of the semester.

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WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR

(An Autonomous Institute)

First Year B. Tech. (Group C), Semester-I

GE1BS2 Engineering Chemistry

Teaching Scheme

Lectures 2 Hours/week

Practical 2 Hours/week

Credits 3

Examination Scheme

ESE 60 Marks

ISE 40 Marks

ICA 25 Marks

Introduction

The Engineering Chemistry course is designed to provide engineering students with foundational knowledge of chemical principles and their practical applications in various engineering domains. It emphasizes the chemistry of fuels, materials, water treatment processes, and analytical techniques crucial to industrial operations. Additionally, the course cultivates awareness of socio-environmental challenges, equipping students with a scientific understanding of global issues like pollution and sustainability. Through theory and numerical applications, the course bridges scientific concepts with real-world engineering problems.

Course Prerequisites

To successfully undertake the Engineering Chemistry course, students should have a basic understanding of chemistry as taught at the higher secondary (10+2) level, particularly in areas such as chemical bonding, physical chemistry, and organic chemistry. Familiarity with fundamental concepts like pH, concentration units, periodic table trends, and basic stoichiometry will aid in grasping advanced topics like fuel chemistry, water treatment, and polymers. A general awareness of environmental issues and industrial processes will further enhance comprehension of socio-environmental applications and analytical techniques introduced in the course.

Course Objectives

1. To impart fundamental knowledge of fossil fuels, their classification, and methods of determining calorific value.
2. To introduce the concepts of engineering materials such as alloys, polymers, plastics, and lubricants, focusing on their properties, uses, and characterization techniques.
3. To familiarize students with water quality parameters, water treatment methods, and their industrial relevance.
4. To provide an understanding of key analytical techniques including chromatography, thermal analysis, and electroplating with practical applications.
5. To raise awareness about major socio-environmental issues such as global warming, acid rain, ozone depletion, and water conservation practices.



Course Outcomes		
At the end of this course, students will be able to: 1. Classify different types of fuels and evaluate their calorific value using theoretical and experimental methods. 2. Describe the composition, properties, and applications of engineering materials like alloys, polymers, and lubricants, and solve related numerical problems. 3. Analyze water quality parameters, calculate hardness, and explain methods for water softening and primary treatment. 4. Apply basic analytical techniques such as chromatography and thermal analysis, and explain the electroplating process and phase rule with applications. 5. Demonstrate understanding of contemporary environmental problems and suggest sustainable solutions like rainwater harvesting.		
Unit – I	Fuels	6 Hours
Fuel: Classification and characteristics of fossil fuels. Comparison between solid, liquid, and gaseous fuels. Determination of the calorific value of fuels: Bomb Calorimeter and Boy's Calorimeter. Numerical: Determination of the calorific value of a fuel sample using Dulong's formula, Bomb Calorimeter and Boy's Calorimeter. Petroleum: Introduction, composition, classification, origin, and refining of crude oil.		
Unit – II	Engineering Materials	6 Hours
Alloys: Definition and purposes of making alloys. Metallic Materials: Properties, composition, and applications of cast iron, steel, and wrought iron. Polymers: Introduction, definitions, polymerization, types of polymerization, and degree of polymerization. Numerical: Degree of polymerization and Number average molecular weight. Plastics: Definition, properties, and types of plastics (Thermosoftening and thermosetting)., Properties and applications of PVC and PET Lubricants: Definition and functions., Characteristic properties (definitions only): viscosity, viscosity index, flash point & fire point, cloud point & pour point, aniline point, oiliness, saponification value, and acid value., Numerical: Calculation of saponification value and acid value.		
Unit – III	Water Parameters and Treatment	6 Hours
Introduction to water quality parameters: pH, acidity, alkalinity, total solids, dissolved oxygen, chlorides, BOD, COD. Hardness of water: Types (temporary and permanent), Numerical: Determination of Hardness. Scale and sludge formation in boilers: causes, disadvantages, and Prevention. Water softening methods: Ion exchange process and Zeolite process. Primary treatment of water: Aeration, sedimentation with coagulation and disinfection.		

Unit – IV	Analytical Techniques	6 Hours
Strength of solution: Definitions and numerical based on Normality, Molarity, and Molality. Chromatography: Definition and types, Gas-Liquid Chromatography (GLC): Definition, instrumentation, and applications. Thermal Analysis: Thermogravimetric Analysis (TGA): Definition, instrumentation, and applications. Electroplating: Process and factors affecting the nature of deposit. Electroplating of copper. Phase Rule: Definition and terminology. Application to one-component system (water): triple point, Metastable systems, and the effect of temperature and pressure on equilibriums.		
Unit – V	Socio-Environmental Problems	6 Hours
Global warming and greenhouse effect. Acid rain and ozone layer depletion. Water conservation techniques with emphasis on rainwater harvesting.		
Internal Continuous Assessment (ICA) ICA consists of performance of eight experiments based on the units. 1. Estimation of Copper in Bronze metal alloy. 2. Determination of Hardness of water. 3. Determination of Alkalinity of water. 4. Determination of Chloride content in given water. 5. Determination of dissolved oxygen in water. 6. Preparation of phenol formaldehyde resin. 7. Preparation of urea Formaldehyde resin. 8. Determination of COD of water sample. 9. Separation of mixture of organic compounds by TLC. 10. Determination of oil and grease from waste water. 11. Hands on Determination of pH by pH meter. 12. Hands on Determination of EMF of acid base titration by potentiometer.		
Text Books		
1. A textbook of Engineering Chemistry, S.S. Dara, S.SUMare, S.Chand Publication 2. A text book of Engineering Chemistry, Shashi Chawala, Dhanpat Rai & Co. 3. A text book of Experiments and Calculations in Engineering Chemistry, S. S. Dara. S. Chand Publication		
Reference Books		
1. Engineering Chemistry, Jain and Jain, Dhanpat Rai & Co. 2. Engineering Chemistry, M. Subha Ramesh, Dr. S. Vairan (Ed.), IInd Wiley 3. Instrumental Methods of Chemical Analysis, Chatwaland Anand, Himalaya Pub. House 4. Industrial Chemistry, B.K. Sharma, Goyal		

5. Chemistry for Engineers, Rajesh Agnihotri, Wiley
6. Fundamentals of Engineering Chemistry, S. K. Singh, New Age Int.
7. Engineering Chemistry, B. L. Tembe, Kamaluddin & M.S. Krishnan, NPTEL Webbook

e-Resources

- <https://youtube.com/playlist?list=PLM-jfaoaU5ixWTj7vaqQF3Ov3FP8yHks5&si=qwQJ0vXWtTUkKJ1v>





WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR

(An Autonomous Institute)

First Year B. Tech. (Group C), Semester-I

GE1BS3 Engineering Mathematics-I

Teaching Scheme

Lectures 2 Hours/week

Tutorial 1 Hour/week

Credits 3

Examination Scheme

ESE 60 Marks

ISE 40 Marks

ICA 25 Marks

Introduction

Engineering Mathematics-I is a fundamental course designed to equip students with essential mathematical tools and techniques necessary for engineering analysis and problem-solving. The course covers key topics such as linear algebra, series expansions, partial differentiation, and complex variable theory. Emphasizing both theoretical understanding and practical applications, it lays the groundwork for advanced mathematical concepts used in various engineering disciplines.

Course Prerequisites

Students enrolling in Engineering Mathematics-I should have a strong foundation in higher secondary mathematics, particularly in algebra, calculus, and coordinate geometry. Familiarity with basic concepts such as functions, limits, derivatives, matrices, and complex numbers is essential, as the course builds on these to explore more advanced topics like eigenvalues, Taylor series, partial differentiation, and complex variable theory. Logical reasoning and algebraic manipulation skills are important for solving systems of equations, analyzing multivariable functions, and constructing analytic expressions relevant to engineering applications.

Course Objectives

1. To introduce students to the concepts of matrix rank, solutions to systems of simultaneous equations, and the determination of eigenvalues and eigenvectors.
2. To introduce students to the expansion of functions and the evaluation of indeterminate forms in limits.
3. To introduce students to the differentiation of functions of several variables and explore its applications.
4. To introduce the tools of differentiation of functions of complex variable that are used in various techniques dealing engineering problems.

Course Outcomes

At the end of this course, students will be able to:

1. Determine the rank of a matrix using echelon form and Solve systems of linear equations.
2. Determine eigenvalues and eigenvectors of real matrices and apply the Cayley–Hamilton theorem.
3. Apply Taylor's and Maclaurin's theorems to find standard series expansions, and evaluate



indeterminate forms using L'Hospital's Rule. 4. Evaluate partial derivatives, apply Euler's theorem, use Jacobians, and optimize multivariable functions. 5. Identify, differentiate, and apply Cauchy-Riemann equations to analyze and construct analytic functions.		
Unit – I	Linear Algebra Part I	5 Hours
Rank of matrix by echelon form, System of Linear Equations-Homogeneous and Non homogeneous equations.		
Unit – II	Linear Algebra Part II	5 Hours
Characteristic equation – Eigen values and Eigen vectors of a real matrix – Properties of Eigen values and Eigen vectors, Cayley – Hamilton theorem (excluding proof)		
Unit – III	Expansion of Functions and Indeterminate forms	5 Hours
Taylor's and Maclaurin's theorem, Standard series expansions, indeterminate forms by L' Hospital Rule.		
Unit – IV	Partial Differentiation	5 Hours
Functions of several variables, Partial Derivatives, Composite Function, Total Derivative, Change of Independent variables, Euler's Theorem on Homogeneous functions		
Unit – V	Applications of Partial Differentiation	5 Hours
Jacobians and its applications, Errors and Approximations, Maxima and Minima of functions of two variable.		
Unit – VI	Complex Variable (Differentiation)	5 Hours
Introduction to complex number, Differentiation, Cauchy Riemann Equations (Without Proof), Analytic Functions, Elementary Analytic Functions (Exponential, Trigonometric and Logarithmic function), Harmonic Functions and Conjugates, Construction of Analytic Function.		
Internal Continuous Assessment (ICA) ICA shall be based on student's performance during tutorial sessions and on completion of six assignments one on each unit.		



Text Books
1. A Text Book of Applied Mathematics, P. N .and J. N. Wartikar, Vol.1, Pune Vidyarthi Griha Prakashan 2. Advanced Engineering Mathematics, H. K. Das, S. Chand Publications, Delhi 3. Engineering Mathematics (Volume), ITL Education, Cengage Learning 4. Engineering Mathematics, Ravish R. Singand, Mukul Bhatt, McGraw-Hill
Reference Books
1. Higher Engineering Mathematics (42nd Edition), B.S. Grewal, Khanna Publications, Delhi 2. Engineering Mathematics, Srimanta Paland, Subodh C. Bhunia, Oxford Higher Education 3. Higher Engineering Mathematics, Ramana B.V., Tata McGraw Hill New Delhi, 2010 4. Applied Mathematics-I, II, Kreyzig's, Wiley
e-Resources
• https://www.youtube.com/playlist?list=PLM-jfaoaU5iwcLA3lrWhIPXCvyZ3JPDbV





WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR
(An Autonomous Institute)
First Year B. Tech. (Group C), Semester-I

EC1ES1 Basic Electrical and Electronics Engineering

Teaching Scheme		Examination Scheme	
Lectures	2 Hours/week	ESE	60 Marks
Practical	2 Hours/week	ISE	40 Marks
Credits	3	ICA	25 Marks

Introduction

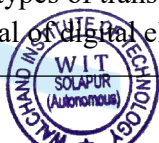
Electrical and Electronic engineers work at the forefront of practical technology, improving the devices and systems we use every day. From solar-energy systems to mobile phones, we innovate to meet society's communication, tech. and energy needs. Electricity is integral to modern life - power generation, transport, medicine, quantum information, and electrical engineering deal with the supply and flow of power; circuits where electricity flows from one point to another, and high-voltage applications with heavy current. Electronic engineering is about electrical circuits and components, creating, designing, and testing them, and integrating them into the hardware of the systems, and circuits that process. Both fields focus on real-world applications.

Course Prerequisites

Before undertaking the Basic Electrical and Electronics Engineering course, students should be familiar with the fundamentals of physics, particularly electricity and magnetism, as introduced at the higher secondary level. A basic understanding of Ohm's law, current, voltage, resistance, and simple circuit elements is essential. Students are also expected to have preliminary exposure to mathematics, including algebra and trigonometry, which are necessary for analyzing AC/DC circuits and solving equations related to electrical quantities. Familiarity with basic binary number systems and logic gates will provide a helpful foundation for the digital electronics portion of the course.

Course Objectives

1. To introduce student various simplification methods for DC circuits so that student can analyze DC circuits and can solve numerical problems based on it
2. To introduce student fundamentals of magnetic circuits and electromagnetic induction and their application.
3. To make student comprehend the generation and behavior of single-phase and poly-phase AC circuits.
4. To make student comprehend the working of a diode and bipolar junction transistor.
5. To make student understand different types of transducers & application areas of transducers.
6. To introduce to student the fundamental of digital electronics.



Course Outcomes		
At the end of this course, students will be able to: 1. Apply the various simplification methods to analyze dc circuits. 2. Use the concept of magnetic circuits to analyze different application parameters. 3. Apply knowledge of AC fundamentals to analyze AC circuits. 4. Explain the working, characteristics, and applications of diode and BJT. 5. Select appropriate transducers to measure various physical parameters like distance, temperature, etc. 6. Perform arithmetic operations on digital number systems and solve Boolean expressions.		
Unit – I	DC Circuits	4 Hours
Kirchhoff's voltage and current law & its implementations, simplifications of resistive circuits - series and parallel combinations, star-delta, delta-star conversions		
Unit – II	Electromagnetism	5 Hours
Series Magnetic Circuits: Introduction, Faraday's law of electromagnetic induction, Lenz's law, self and mutual inductance. Single-Phase Transformer: Working principle and construction. DC motor: Construction, working principle, and speed control.		
Unit – III	AC Circuits	6 Hours
Single phase AC: Generation of voltage and current, RMS value, average value, form factor, crest factor, and peak factor. AC circuits: R, L and C circuits, its behavior for single phase AC and phasor diagram. Three phase AC: Generation, star and delta connections for balanced systems.		
Unit – IV	Diodes and Rectifiers	4 Hours
P-N junction diode: Construction, working, V-I characteristics. Rectifiers: Circuit diagram & working of half wave rectifier, full wave rectifier, bridge rectifier, bridge rectifier with capacitor filter. Special Purpose Diodes: Photodiode, Zener, LED-working and application		
Unit – V	Bipolar Junction Transistor	3 Hours
BJT: Construction, working, current components, relationship between α and β, CB, CE, CC configuration, I/O characteristics of CE configuration. Application of BJT: Switch and CE amplifier.		



Unit – VI	Electrical Transducers	2 Hours
Parameters for selection of transducers. Construction and Working: Strain gauge, LVDT, thermocouple, thermistor, LDR, solar cell.		
Unit – VII	Digital Electronics	6 Hours
Number system: Decimal, binary, octal, hexadecimal & their inter-conversion, BCD code Binary Arithmetic: Addition, subtraction, subtraction using 2's complement Logic Gates: Symbol, output equation, truth table, the realization of basic gates using universal gates. Boolean Algebra: Laws & rules, De- Morgan theorem		
Internal Continuous Assessment (ICA) ICA consists of the performance of eight experiments based on the units.		
Text Books		
1. Electrical Technology (Volume I & 2), B L Theraja, 22nd edition, S. Chand & Company Ltd. 2. Basic Electrical Engineering, V K Mehta, Revised edition, S. Chand & Company Ltd. 3. Basic Electronics Solid State, B L Thereja, Revised edition, S. Chand & Company Ltd.		
Reference Books		
1. Electrical Technology, E. Hughes, 10th edition, ELBS, Longman 2. Electronic Devices and Circuits, David A. Bell Oxford University, Press India, Fifth edition 3. Principles of Electronic Devices and Circuits (Analog and Digital), B.L. Theraja, R. S. Sedha, S. Chand Publication 4. Digital Principles and Applications, Albert Malvino, Donald Leach, Tata McGraw Hills Publication		
e-Resources		
• https://youtube.com/playlist?list=PLM-jfaoaU5ixvQUWrSTYex62TUx0hOkEb&si=w9CnJC_hqezo72fq		





WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR

(An Autonomous Institute)

First Year B. Tech. (Group C), Semester-I

GE1AE1 Professional & Communication Skills

Teaching Scheme

Lectures 2 Hours/week

Tutorial 1 Hour/week

Credits 3

Examination Scheme

ESE 60 Marks

ISE 40 Marks

ICA 25 Marks

Introduction

In today's globalized and technologically advanced world, effective communication is a vital professional skill for engineering graduates. This course, is designed to strengthen students' command of the English language, develop technical communication proficiency, and build essential employability skills. Through a blend of grammar, verbal aptitude, technical writing, listening, reading, and speaking activities, the course aims to make students industry-ready communicators equipped to perform in diverse professional scenarios.

Course Prerequisites

Students enrolling in the Communication Skills course should have a basic understanding of English language fundamentals, as typically acquired during school education. They are expected to possess elementary proficiency in grammar, vocabulary, reading, and writing. While fluency is not essential at entry, a willingness to participate actively in listening and speaking exercises, engage in group discussions, and practice technical writing is important. Students should also be open to improving their interpersonal and professional communication, as the course emphasizes real-world applications relevant to workplace and academic environments.

Course Objectives

1. Introduce students to the fundamental principles, types, and barriers of communication, while developing effective interpersonal communication strategies.
2. Strengthen students' command of English grammar, sentence structure, vocabulary, and error identification for technical and professional communication.
3. Develop effective listening and speaking skills for formal and informal technical contexts, including presentations and group discussions.
4. Foster students' abilities to read, interpret, and critically analyze technical and professional content such as articles, reports, and case studies.
5. Enhance professional writing skills through practice in composing clear, concise, and well-structured documents, including emails and reports relevant to industry and academic settings.



Course Outcomes		
At the end of this course, students will be able to: 1. Understand the communication process and apply effective strategies to overcome barriers and enhance interpersonal communication 2. Strengthen verbal and grammatical skills in professional and technical settings. 3. Demonstrate participative listening and speaking skills for technical and workplace scenarios using proper etiquette and fluency. 4. Develop critical reading and analytical abilities to comprehend, interpret, and evaluate technical and professional texts. 5. Compose well structured professional documents such as emails and reports, demonstrating proficiency in written communication.		
Unit – I	Introduction to Communication Skills	3 Hours
Communication, Process of Communication, Principles of Effective Communication, Types of Communication: Verbal, Non-Verbal, formal, informal, Barriers to Communication, 7Cs of Effective Communication, Effective measures to improve interpersonal communication skills.		
Unit – II	Verbal Ability Skills	8 Hours
Parts of Speech, Kinds of Sentences – Identification and Transformation, Sentence Pattern – Framing Sentences, Tenses – Rules and its usage, Vocabulary: Word Formation: Prefixes, Suffixes, Root words, idioms& phrases, Common Errors, Sentence Correction, Sentence Rearrangement, Verbal Analogy, and Spot the Error.		
Unit – III	Listening and Speaking Skills	6 Hours
Application of different types of listening and speaking, -Active, Passive, Critical, Empathetic, Selective Listening, Informative, Persuasive, Demonstrative, Extempore Speaking. Situational Conversation, Telephonic Etiquettes, Role Play, Extempore Speech and Public speaking.		
Unit – IV	Reading Skills	6 Hours
Application of different types of Reading: - Skimming, Scanning, Intensive and Extensive Reading, Reading Comprehension, reading and Evaluating Start-Up Case Studies.		
Unit – V	Writing Skills	6 Hours
Application of different types of Writing: Drafting notices, cover letter, Effective e-mail communication, paragraph, essay writing and creative writing.		



Internal Continuous Assessment (ICA)

Students must complete at least 10 exercises from the list below during lab and tutorial sessions:

1. Evaluate messages using the 7 Cs of communication
2. Classify words by parts of speech
3. Transform sentences (e.g. assertive to interrogative)
4. Build vocabulary using prefixes, suffixes, and root words
5. Spot and correct grammatical errors
6. Listen actively to audio clips and answer questions
7. Participate in formal and informal situational conversations (role-plays)
8. Deliver extempore speeches on assigned topics
9. Practice telephonic etiquette through role-plays
10. Making Creative advertisement and Storytelling
11. Apply skimming and scanning techniques on technical texts
12. Draft a notice, cover letter and emails on technical topics
13. Compose paragraphs and essays related to engineering

Text Books

1. Business Communication, Shalini Kalia, Shailja Agarwal, Wiley
2. Communication Skills for Technical Students, T. M. Farhathullah, Orient Black Swan, 2003
3. English Grammar Just for You, Rajeevan Karal, Oxford University Press
4. English for Practical Purposes, Z. N. Patil, B.S. Valke, A.R. Thorat, Zeenath Merchant
5. Study Writing, Liz Hamp-Lyons and Ben Heasley, Cambridge University Press, 2006
6. Exercises in Spoken English, Parts I-III, CIEFL, Hyderabad, Oxford University Press
7. Communication Skills, Sanjay Kumar and Pushpa Lata, Oxford University Press, 2011
8. The Functional Aspects of Communication Skills, Dr. P. Prasad, New Delhi
9. Soft Skills: A Textbook for Undergraduates, Ajay R. Tengse, Orient Black Swan
10. Business Communication, Nawal, Cengage Publication: 9789353502157
11. Soft Skills for Everyone, Butterfield, Cengage Publication: 9789353501051
12. Technical English: Course Book, Terry Philips, Garnet Education, 2011

Reference Books

1. English Grammar & Composition, Wren & Martin, S. Chand Publication, 2015
2. Practical English Usage, Michael Swan, OUP, 1995
3. Remedial English Grammar, F.T Wood, Macmillan, 2007
4. On Writing Well, William Zinsser, Harper Resource Book, 2001
5. Longman Dictionary of Contemporary English, Longman, Harlow, England, 1978
6. Essential Activator, Longman, Harlow, 2006
7. Word Power Made Easy, Norman Lewis, 2017
8. Communicative English for engineers and professionals, B. Nitin, Pearson Education India, 2010
9. Business communication for success, S. McLean, 2010
10. English Language Communication Skills. Lab Manual cum Workbook JNTU Board of Editors.

- | |
|--|
| 11. Fusion: Integrated Reading and Writing, Book 1, Kemper, Meyer, Van Rys, Sebranek
12. Handbook of Professional, Business & Technical Writing, and Communication and Journalism: A Reference Guide to All Kinds of Writing, 2023
13. English Grammar Just for You, Rajeevan Karal, Oxford University Press |
| e-Resources |
| <ul style="list-style-type: none">• https://youtube.com/playlist?list=PLM-jfaoaU5iyQgWy-WHgFc2G21J-agChw&si=kaCVsm5MJ-dSmtsT |





WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR
(An Autonomous Institute)
First Year B. Tech. (Group C), Semester-I

GE1IK1 Indian Science & Technology

Teaching Scheme		Examination Scheme	
Lectures	2 Hours/week	ESE	--
Tutorial	--	ISE	25 Marks
Credits	2	ICA	25 Marks

Introduction

The course Indian Science & Technology explores the rich heritage of scientific knowledge and technological practices developed in India through the ages. Spanning subjects from metallurgy and textile chemistry to water management, astronomy, and environmental planning, the course highlights the innovative and sustainable methods employed by Indian civilizations. It provides an interdisciplinary perspective, connecting traditional scientific achievements to their relevance in modern contexts, while fostering appreciation for indigenous knowledge systems and their role in shaping technological progress.

Course Prerequisites

To fully benefit from the Indian Science & Technology course, students should possess a general awareness of Indian history and basic school-level understanding of scientific concepts across disciplines such as physics, chemistry, and environmental science. An interest in cultural heritage, traditional knowledge systems, and historical innovations will enrich their learning experience. Familiarity with interpreting historical texts or archaeological findings, although not mandatory, can aid in grasping the context of India's scientific achievements and their modern-day relevance.

Course Objectives

1. To provide an overview of India's historical contributions to science and technology across various domains.
2. To explore traditional Indian advancements in metallurgy, textile chemistry, and ceramic technologies.
3. To explore indigenous water management systems, transportation methods, and maritime developments.
4. To study key Indian contributions in mathematics and astronomy, including the evolution of calculus and time measurement.
5. To understand traditional ecological and environmental practices such as agroforestry, vernacular architecture, and water conservation.



Course Outcomes		
At the end of this course, students will be able to: 1. Recognize the historical evolution and cultural richness of India's scientific and technological heritage. 2. Explore the scientific principles underlying traditional Indian practices in metallurgy, textile chemistry, and pyro technology, and evaluate their relevance in the modern context. 3. Interpret indigenous methods of water management and transportation, and analyze their sustainable and innovative features. 4. Identify Interpret India's contributions to mathematics, astronomy, ecology, and environmental practices, and assess their impact on global scientific advancements.		
Unit – I	Fundamentals	4 Hours
• An overview of Indian contributions to technology • Technological Innovations		
Unit – II	Metallurgy, Textile Chemistry & Pyro Technology	5 Hours
• Copper/Bronze/Zinc: Important Mines (Zawar, Khetri mines) • Iron and Wootz Steel Technology • Textile and Dyeing – Indian Specialities (Kutchi Embroidery, Cotton Textile, etc.) • Ceramic Technology, Stone (Lapidary), Shell, Ivory, Faience & Glass Technology		
Unit – III	Water Management & Transportation	5 Hours
• Harappan and Traditional Water Management System of Gujarat • Historical Sites – Sringaverpur, South Indian Water Management System, Western Ghats Cave Kanheri, etc. • Communities Involved in Water Management • Modes of Transportation and Reforms • Grand Trunk Road (Uttarapath & Dakshinapath) • Development of Trading Techniques • Boat & Ship Building		
Unit – IV	Mathematics & Astronomy	5 Hours
• Mathematics contained in the Sulba Sutra • Weaving Mathematics into Beautiful Poetry – Bhaskaracharya • The Evolution of Sine Function in India • The Discovery of Calculus by Kerala Astronomers • Vedanga Jyotish & Measuring Time & Calendar		

Unit – V	Ecology and Environment	5 Hours
• Nakshatra Gyaan and Agriculture • Vernacular Architecture • Forest Management and Urban Planning • Agroforestry • Tanks, Lakes, and Stepwells		
Unit – VI	India's Contribution in Science and Technology to the World	4 Hours
India's Contribution in Science and Technology to the World		
Internal Continuous Assessment (ICA) ICA shall be based on completion of Five Assignments each topic mentioned above.		
Text Books		
1. Introduction to Indian Knowledge System: Concepts and Applications, Mahadevan B., Bhat, Vinayak Rajat & Nagendra, Pavana R. N., PHI Learning Pvt. Ltd., 2022		
Reference Books		
1. Pride of India: A Glimpse into India's Scientific Heritage, R. M. Pujari, Pradeep Kolhe, N. R. Kumar, Samskrita Bharati Publication 2. Indian Contribution to Science, compiled by Vijnana Bharati 3. Knowledge Traditions and Practices of India, Kapil Kapoor, Michel Danino, CBSE, India 4. Dr. Subhash Kak, Computation in Ancient India, Mount Meru Publishing, 2016 5. Indian Science and Technology in the Eighteenth Century, Dharampal, Academy of Gandhian Studies, Hyderabad, 1971, republished by Other India Bookstore, Goa, 2000 6. The Man Who Knew Infinity: A Life of the Genius Ramanujan, Robert Kanigel, Abacus, London, 1999 7. Sciences of the Ancient Hindus: Unlocking Nature in the Pursuit of Salvation, Alok Kumar, Create Space Independent Publishing, 2014 8. Science in India: A Historical Perspective, B. V. Subbarayappa, Rupa, New Delhi, 2013 9. Indian Mathematics and Astronomy: Some Landmarks, S. Balachandra Rao, Jnana Deep Publications, Bangalore, 3rd edn, 2004 10. Vedic Mathematics and Science in Vedas, S. Balachandra Rao, Navakarnataka Publications, Bengaluru, 2019 11. Ancient Hindu Geometry: The Science of the Śulba, Bibhutibhushan Datta, 1932, Repr., Cosmo Publications, New Delhi, 1993 12. History of Hindu Mathematics, Bibhutibhushan Datta & Avadhesh Narayan Singh, 1935, Repr., Bharatiya Kala Prakashan, Delhi, 2004 13. The Crest of the Peacock, George G. Joseph, Penguin Books, London & New Delhi, 2000		

14. Ancient Cities, Sacred Skies: Cosmic Geometries and City Planning in Ancient India, J. McKim Malville & Lalit M. Gujral, IGNC&A & Aryan Books International, New Delhi, 2000
15. Chasing Shadows: Mathematics, Astronomy and the Early History of Eclipse Reckoning, Clemency Montelle, Johns Hopkins University Press, 2011
16. Jantar Mantar: Maharaja Sawai Jai Singh's Observatory in Delhi, Anisha Shekhar Mukherji, AMBI Knowledge Resources, New Delhi, 2010
17. Astronomy in India: A Historical Perspective, Thanu Padmanabhan (ed.), Indian National Science Academy, New Delhi & Springer (India), 2010
18. A History of Hindu Chemistry, Acharya Prafulla Chandra Ray, 1902, Republ., Shaibya Prakashan Bibhag, centenary edition, Kolkata, 2002
19. Delhi Iron Pillar: New Insights, R. Balasubramaniam, Indian Institute of Advanced Study, Shimla & Aryan Books International, New Delhi, 2002
20. Marvels of Indian Iron Through the Ages, R. Balasubramaniam, Rupa & Infinity Foundation, New Delhi, 2008
21. Dying Wisdom: Rise, Fall and Potential of India's Traditional Water-Harvesting Systems, Anil Agarwal & Sunita Narain (eds.), Centre for Science and Environment, New Delhi, 1997
22. The Iconography of Water: Well and Tank Forms of the Indian Subcontinent, Fredrick W. Bunce, D K Printworld, New Delhi, 2013

e-Resources

- <https://iksindia.org/>





WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR
(An Autonomous Institute)
First Year B. Tech. (Group C), Semester-I

GE1CC1N* Co-curricular Activity

Teaching Scheme		Examination Scheme	
Lectures	--	ESE	--
Tutorial	--	ISE	--
Credits	1	ICA	25 Marks

Introduction

Students are required to go through the list of following Co-curricular Activities and select any one of their interests. On campus/Online activities, discussions, presentations and lecture methods will be conducted by experts from respective activity.

Students are required to submit hard copy of a report on the activities performed related to topics opted for Co-curricular Activity. Evaluation will be done based on the report submitted by student.

Selecting co-curricular activities that align with your interests and goals can significantly enrich your educational journey. Remember to maintain a balance and choose activities that you are genuinely excited about. This approach will help you gain the most from your co-curricular activities.

Course Objectives

1. To promote overall physical, mental, and emotional well-being through yoga practices, sports activities, and healthy lifestyle habits.
2. To develop discipline, teamwork, stress management, and awareness about fitness, nutrition, and personal health.

Course Outcomes

At the end of this course, students will be able to:

1. Apply yoga techniques, sports skills, and healthy practices to maintain physical fitness and a healthy lifestyle.
2. Demonstrate improved physical fitness, mental balance, teamwork, and a positive attitude towards overall well-being through participation in sports and wellness activities.

Co-Curricular Course

1. Yoga (GE1CC11)
2. Health and Wellness (GE1CC12)
3. Sports (GE1CC13)



Study Material

1. Yoga
 - https://onlinecourses.swayam2.ac.in/aic22_ge09/preview
 - https://onlinecourses.swayam2.ac.in/cec25_ed19/preview
 - https://onlinecourses.nptel.ac.in/noc25_mg104/preview
 - https://onlinecourses.nptel.ac.in/noc21_hs29/preview
 - https://www.youtube.com/playlist?list=PLZF4IOyPf_Z9yQ_Gphjh1wZ5fEhMg6hmP
2. Health and Wellness
 - https://onlinecourses.nptel.ac.in/noc25_hs109/preview
 - https://onlinecourses.nptel.ac.in/noc21_hs29/preview
3. Sports
 - https://onlinecourses.nptel.ac.in/noc25_hs156/preview





WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR

(An Autonomous Institute)

First Year B. Tech. (Group C), Semester-I

CS1PC1L C Programming

Teaching Scheme		Examination Scheme	
Lectures	2 Hours/week	ESE	50 Marks
Practical	2 Hours/week	ISE	25 Marks
Credits	3	ICA	25 Marks

Introduction

C programming language is a procedural and general-purpose programming language. It is fast and simple to learn and implement. It includes low-level memory access and simple syntax. It makes the C programming language suitable for system programming like compiler development and operating system development. C language is also known as a structured programming language. C language has features like simple, powerful, structure-oriented, case-sensitive, fast and efficient.

Course Prerequisites

To effectively learn C Programming, students should possess a basic understanding of computers, including familiarity with using an operating system and basic typing skills. Logical reasoning and problem-solving abilities, such as identifying steps in a process or solving simple puzzles, are essential for grasping algorithms and flowcharts. Comfort with basic mathematical concepts like arithmetic operations, variables, and order of operations will aid in learning expressions and control structures. Exposure to general programming ideas or block-based coding (like Scratch or flow-based logic) is beneficial but not mandatory. Overall, an analytical mindset and foundational computational thinking are important prerequisites for this course.

Course Objectives

1. To introduce the concepts of algorithms & flowcharts for problem solving.
2. To introduce the fundamentals of structured programming.
3. To formulate simple C programs using various control structures & loop structures.
4. To apply the concept of functions & pointers to solve the problems.
5. To apply the concept of arrays to solve the problems.
6. To apply the concept of structures, unions and files to solve the problems.

Course Outcomes

At the end of this course, students will be able to:

1. Design the flowcharts and algorithms for the given problem.
2. Translate the algorithms into C programs and test & execute the programs.
3. Design & develop C programs by appropriately selecting control and loop structures.
4. Design & develop C programs using functions and pointers.
5. Design & develop C programs using arrays, structures, unions and files.

Unit – I	Fundamentals of Computers & Structure of a 'C' Program	6 Hours
Techniques for Problem Solving: algorithm, flow chart, examples. Algorithms specifications, formulation of simple algorithms and logical problems. Structure of C program, building blocks of C, program (preprocessor, compilation and execution & debugging of C program), IDE, C character set, tokens, constants, variables, keywords, primitive data types, 'C' operators (arithmetic, unary, binary, ternary, Logical, assignment, relational, increment and decrement, conditional, bitwise, size of) Operator precedence, expressions, typecasting and type conversion, formatting input and output (getchar, putchar, printf, scanf).		
Unit – II	Control Structures	5 Hours
Control Statement: if, if-else, nested if –else, else if ladder Loops: while, do-while, for, nesting of loops, break, continue, goto statement, switch-case statement.		
Unit – III	Arrays and Strings	5 Hours
Declaration and initialization of one-dimensional array, accessing elements, Array handling. String handling functions. Declaration and initialization of two-dimensional arrays, accessing elements, Array handling, Matrix operations		
Unit – IV	Structures and Unions	3 Hours
Definition of structure and union, declaration, Accessing elements and displaying elements, difference between structure and union.		
Unit – V	Pointers	3 Hours
Introduction, Declaration of pointer, initialization, accessing pointer, pointer to basic data types, pointer to array (one dimensional)		
Unit – VI	Functions	6 Hours
Declaration & definition, passing parameters to functions, pass by value, scope of variable, return statement. Function using call by value and call by reference Recursion: Processes and Recursion in C. How recursion works, Factorial, Fibonacci sequence, Towers of Hanoi, Advantages and Disadvantages of recursive Techniques		
Unit – VII	Files	3 Hours
Introduction, Streams and file types, File operations, File I/O, Structures of read and write.		



Internal Continuous Assessment (ICA)

ICA shall be based on the programming assignments.

• List of Assignments:

1. Fundamentals of algorithm & flow chart, structure of a C program
2. Formatted input/output functions and different operators in C
3. Control structures - Decision control statement (if, if-else, nested if-else, else if ladder), Loop Statement (for, while, do-while, nested loop) and Switch-case statement
4. Arrays(one-dimensional & two-dimensional) and string operations
5. Structures and unions
6. Pointers and dynamic memory allocation
7. Functions (call-by-value & call-by-reference), recursion
8. File operations

POE (#): The Course 'Programming for Problem Solving' shall have a 'Practical and Oral Examination' at the end of the semester assessing student's programming

Text Books

1. Schaum's Outline of Programming with C, Byron Gottfried, Mc Graw-Hill
2. Simplifying C, Harshal Arolkar & Sonal Jain, Dreamtech (For Unit-01)
3. Lets 'C', Yaswant Kanetkar, BPB Publication
4. Programming in ANSIC, C. Balgurusamy, Tata Mc GrawHill

Reference Books

1. The C Programming Language (ANSIC Version), Brian W. Kernighan, Dennis M. Ritchie, PHI Publications
2. Programming in C, B. L. Juneja, Cengage Learning
3. Projects using C, P. V. N. Varalaxmi, Scitech Publications
4. The C Programming Language, Brian W. Kernighan and Dennis M. Ritchie, Prentice Hall of India

e-Resources

- <https://www.w3resource.com/c-programming-exercises/>
- <https://www.geeksforgeeks.org/c-programming-language/>
- https://www.tutorialspoint.com/cprogramming/c_useful_resources.htm
- <https://www.learn-c.org/>





WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR
(An Autonomous Institute)
First Year B. Tech. (Group C), Semester-I

GE1CC2L Creativity and Design Thinking

Teaching Scheme		Examination Scheme	
Lectures	1 Hour/week	ESE	--
Practical	2 Hours/week	ISE	--
Credits	2	ICA	50 Marks

Introduction

Creativity and Design Thinking is an interdisciplinary course that empowers students to solve problems innovatively through user-centered approaches. The course begins by exploring memory and cognitive processes that influence creative thinking and gradually introduces structured methods such as brainstorming, lateral thinking, and TRIZ. Students learn to navigate the five-stage design thinking process—Empathize, Define, Ideate, Prototype, and Test—using practical tools and case studies. The focus on feedback and iterative improvement fosters a mindset geared toward continuous innovation and user-focused design.

Course Prerequisites

To successfully engage with the Creativity and Design Thinking course, students should have a basic understanding of cognitive processes and an interest in innovative thinking and human-centered problem-solving. Familiarity with brainstorming methods, basic observation skills, and an openness to iterative learning will support their grasp of the design thinking process. While no advanced technical knowledge is required, the ability to reflect on user experiences, communicate ideas clearly, and work collaboratively in teams will greatly enhance the learning experience.

Course Objectives

1. To relate and compare the various memory techniques and apply them in engineering education.
2. To appreciate the importance of creativity and design thinking.
3. To analyze emotional experience and experiment with emotional expressivity to understand users in a better way, while designing the products.
4. To transform thought process so as to come up with multiple solution using ideation and creative tools.
5. To develop skills for evaluating, articulating, refining, testing and creating an innovative engineering product/prototype that solves customer problems(s).



Course Outcomes		
At the end of this course, students will be able to: 1. Relate and Compare the various memory techniques and apply them in engineering education. 2. Appreciate the importance of creativity and design thinking 3. Analyze emotional experience and Experiment with emotional expressivity to understand users in a better way, while designing the products. 4. Transform their thought process so as to come up with multiple solution using Ideation and creative tools. 5. Develop skills for evaluating, articulating, refining, testing and creating an innovative engineering product/prototype that solves customer problems(s).		
Unit – I	An Insight into Remembering	2 Hours
Understanding the Memory process, Memory enhancement techniques.		
Unit – II	Introduction to Creativity and Tools for creativity	4 Hours
The creative person, Lateral & vertical thinking concept, Creative style- adaptor & Innovator. Tools for creativity: Brain storming, Six hat technique, Introduction to TRIZ, Divergent and Convergent thinking.		
Unit – III	Basics of Design Thinking	4 Hours
Definition of Design Thinking, Need for Design Thinking, Objective of Design Thinking, Stages of Design Thinking Process (explain with examples) – Empathize, Define, Ideate, Prototype, Test, User Persona		
Unit – IV	Process of Empathize	4 Hours
Empathize, Different techniques used, case studies.		
Unit – V	Ideate	4 Hours
Ideate – Explanation with case studies, tools to be used		
Unit – VI	Prototype	2 Hours
Prototype –Explanation with case studies. tools to be used,		
Unit – VII	Testing the Prototype	4 Hours
Testing for prototype, case studies.		



Unit – VIII	Feedback, Re-Design & Re-Crete	4 Hours
Feedback loop, Focus on User Experience, User focused design, rapid prototyping & testing, final product.		
Internal Continuous Assessment (ICA) Activities are to be conducted using various tools of creative and design thinking. Typical case studies shall be done to understand process of design thinking and product development. Assessment shall be carried out on final prototype / product for end user. In addition to the above, the institute may prescribe additional modes of assessment such as product development, Case study, Quiz, Presentation, Course seminar etc. for ensuring continuous assessment of the students.		
Text Books		
1. Thinking Design, S. Balaram, Publisher: Sage India; ISBN-10 :8132103149 2. The Creative thinker's Toolkit: Course Guidebook, Gerard Puccio, The great courses, 2014 3. Six Thinking Hat, Marcela Pandolfo 4. Basic Design Thinking, Gavin Ambrose & Paul Harris, AVA Publishing 5. Design Thinking for Beginners: Innovation as a factor for entrepreneurial success, Publisher : Personal Growth Hackers (18 August 2019), ISBN-10 : 3967160629		
Reference Books		
1. Design Thinking: Understand – Improve – Apply, Hasso Plattner, Christoph Meinel and Larry Leifer (eds), Springer, 2011 (Unit I) 2. Design Thinking for Strategic Innovation: What They Can't Teach You at Business or Design School, Idris Mootee, John Wiley & Sons, 2013 (Unit III) 3. Design Thinking for Educators: Unleashing Imagination Ideas Being Student Centric, D.M., Arvind Mallik, Publisher : Notion Press; 1st edition, 17 September 2019, ISBN-10 : 1646506928 4. Cracking the Creativity code: Zoom in Zoom out framework for creativity, fun and success, Arie Ruttenberg and Shlimo Maital, Publisher- SAGE, 2014		
e-Resources		
1. https://dschool.stanford.edu/.../design resources/.../ModeGuideBOOTCAMP2010L.pdf 2. https://dschool.stanford.edu/use-our-methods/ 3. https://www.interaction-design.org/literature/article/5-stages-in-the-design-thinking- process 4. http://www.creativityatwork.com/design-thinking-strategy-for-innovation/ 5. https://www.nngroup.com/articles/design-thinking/ 6. https://designthinkingforeducators.com/design-thinking/ 7. www.designthinkingformobility.org/wp-content/.../10/NapkinPitch_Worksheet.pdf 8. https://www.behance.net/joblist?ssid=V6WZKQ8K&mv=search		



WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR

(An Autonomous Institute)

First Year B. Tech. (Group C), Semester-II

GE2BS1 Engineering Physics

Teaching Scheme

Lectures	2 Hours/week
Practical	2 Hours/week
Credits	3

Examination Scheme

ESE	60 Marks
ISE	40 Marks
ICA	25 Marks

Introduction

Engineering Physics forms the foundational framework for understanding the physical principles underlying modern engineering and technology. This course explores essential topics including crystallography, semiconductor physics, wave optics, laser and fiber optics, quantum mechanics, nanoscience, nuclear physics, and sound engineering. Emphasizing both theoretical concepts and practical applications, it prepares students to analyze physical phenomena critical to the design and development of advanced engineering systems.

Course Prerequisites

Students taking Engineering Physics should have a solid grounding in higher secondary-level physics and mathematics, especially in topics like waves, optics, atomic structure, and basic calculus. Familiarity with physical concepts such as diffraction, semiconductors, and energy forms will help in understanding advanced topics like quantum mechanics, lasers, and nuclear physics. Basic numerical problem-solving skills are expected, along with an interest in applying physics to real-world technologies such as fiber optics, nanoscience, and acoustics.

Course Objectives

1. To enable students to analyze crystal structures using fundamental concepts and classify solids based on their band gap energy and corresponding electronic properties.
2. To equip students to apply the principles of diffraction and polarization for calculating resolving power and determining specific rotation in optical systems.
3. To ensure students can analyze the working mechanisms of lasers and optical fibers, and articulate their practical applications.
4. To help students interpret wave-particle duality through de Broglie theory and categorize nanomaterials based on their applications.
5. To develop students' ability to critically analyze nuclear fission reactions, compare them with fusion reactions, and to describe the essential components of a nuclear fission reactor.
6. To enable students to evaluate the acoustic performance of enclosed spaces and summarize the methods and applications of ultrasonic wave generation.



Course Outcomes		
After completing the course, students will be able to: 1. Analyze crystal structures using Miller indices & Bragg's law and differentiate solids based on band gap energy and their associated electronic properties. 2. Apply diffraction and polarization principles to calculate resolving power and determine specific rotation in optical systems. 3. Analyze working mechanism of lasers and optical fibers along with their applications in different domains. 4. Interpret wave-particle duality from de Broglie theory and classify nanomaterials used in various fields. 5. Analyze and evaluate nuclear fission reaction, compare it with fusion reaction and describe the essential components of nuclear fission reactor. 6. Evaluate acoustic performance in enclosed spaces and summarize ultrasonic wave production methods and applications.		
Unit – I	Crystallography & Semiconductor Physics	5 Hours
Lattice Planes and Miller Indices, Interplanar Spacing for Cubic System, Symmetry elements (axis, center and plane), Bragg's Law, Numericals. Fermi-Dirac probability distribution function (introduction only), Fermi level in intrinsic and extrinsic semiconductors, Effect of impurity concentration on Fermi level, Derivation for E_{Fin}, Hall effect and applications, Numericals.		
Unit – II	Wave Optics	5 Hours
Diffraction: Introduction, resolving power (R.P.), Rayleigh criterion, theory of diffraction grating and its R.P., Numericals. Polarization: Introduction, Optical activity, Specific rotation, Numericals.		
Unit – III	LASER and Fibre Optics	5 Hours
LASER: Introduction, Interaction of radiation with matter, Population Inversion and Pumping, Characteristics of laser, He-Ne LASER, Applications of LASER. Fibre Optics: Introduction, Structure of optical fibre, Working principle of optical fibre, Acceptance Angle, Numerical Aperture, Advantages and applications of optical fibres, Communication system: basic building blocks, Numericals.		
Unit – IV	Introduction to Quantum Mechanics and Nanoscience	5 Hours
Introduction to Quantum Mechanics: de Broglie hypothesis, de Broglie wavelength of matter waves: in terms of kinetic energy and associated with particle in thermal equilibrium, Properties of matter waves, Davisson-Germer Experiment (apparatus, investigations and analysis), Numericals. Introduction to Nanoscience: Introduction, Types of Carbon Nanotubes, Applications of nanomaterials in different fields.		

Unit – V	Nuclear Physics	5 Hours
Nuclear fission Reaction as a source of energy, chain reaction and multiplication factor (Introduction). Essentials of nuclear reactor, Thermonuclear reactions and types, Difference between Fusion and Fission, Numericals.		
Unit – VI	Sound Engineering	5 Hours
Acoustics: Introduction, Reverberation, Reverberation time, absorption coefficient (definition only), Sabine's formula, Factors affecting acoustics of auditorium and their remedies. Ultrasonic: Introduction, Production methods of ultrasonic waves (piezoelectric effect and magnetostriction effect (introduction only)), Properties of ultrasonic waves and applications, Numericals.		
Internal Continuous Assessment (ICA) ICA consists of performance of eight experiments based on the units. - List of Experiments - XRD analysis - Crystallography - Hall Effect - Band Gap Energy - Theory of Plane Diffraction Grating - Resolving Power of Plane Diffraction Grating - He-Ne Laser: Beam Divergence - Numerical aperture and Acceptance angle of an optical fibre. - Plank's constant - Kundt's Tube		
Text Books		
- A Text Book of Engineering Physics, M.N. Avadhanulu, P.G. Kshirsagar, S. Chand Publications - A Text Book of Optics, Subramanya and Brij Lal, S. Chand and Company Ltd. - Engineering Physics, R. K. Gaur, S. L Gupta, Eighth Revised Edition 2012, Dhanpat Rai Publications (P) Ltd.		
Reference Books		
- Modern Physics, B.L. Theraja, S. Chand and Company Ltd. - Nanotechnology: Principles and Practices, Sulabha K. Kulkarni, Capital Publishing Company - Nanoscience and Nanotechnology: Fundamentals to Frontiers, M.S. Ramachandra Rao, Shubra Singh, Wiley India Pvt. Ltd., New Delhi - Engineering Physics, D.K. Bhattacharya, Poonam Tandon, Oxford University Press - Solid State Physics, S.O. Pillai, McGraw Hill Publications. - An Introduction to Laser's Theory and Applications, Dr. M. N. Avadhanulu, Dr. P. S. Hemne, Revised Edition 2017, S. Chand & Company Pvt. Ltd.		

e-Resources

- <https://youtube.com/playlist?list=PLM-jfaoaU5iz5hfLQrddV8Eb3maIWGCNQ&si=LMyu53kTDIPtkeTu>
- <https://archive.nptel.ac.in/courses/115/106/115106127/>





WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR

(An Autonomous Institute)

First Year B. Tech. (Group C), Semester-II

GE2BS4 Engineering Mathematics-II

Teaching Scheme		Examination Scheme	
Lectures	2 Hours/week	ESE	60 Marks
Tutorial	1 Hour/week	ISE	40 Marks
Credits	3	ICA	25 Marks

Introduction

Engineering Mathematics-II builds upon foundational calculus and differential equations, equipping students with analytical tools for solving complex engineering problems. This course explores first-order differential equations, their applications in physical systems, and numerical methods for their solution. It also introduces integral calculus, multiple integrals, and vector calculus, emphasizing real-world applications in physics, engineering, and computer simulations.

Course Prerequisites

To effectively engage with Engineering Mathematics-II, students should have a solid grasp of basic calculus and algebra as taught in Engineering Mathematics-I or equivalent first-semester coursework. This includes familiarity with differentiation and integration techniques, basic knowledge of coordinate geometry, and an understanding of elementary functions. Additionally, prior exposure to the concepts of limits, continuity, and basic differential equations is essential. Comfort with mathematical reasoning and manipulation will aid in tackling topics such as vector calculus, multiple integrals, and numerical methods introduced in this course.

Course Objectives

1. To introduce students to the methods for solving ordinary differential equations and to demonstrate their applications in engineering and physical sciences.
2. To introduce students to numerical methods for solving ordinary differential equations and to develop their ability to apply these techniques to practical problems.
3. To introduce students to integral calculus and enable them to evaluate improper integrals using special functions.
4. To introduce students to the concepts of gradient, directional derivatives, curl, and divergence, and to explain their significance in vector calculus.

Course Outcomes

At the end of this course, students will be able to:

1. Solve the ordinary differential equations and apply their applications in various engineering problems.
2. Use numerical method for solving the ordinary differential equations.



3. Evaluate the improper integrals by some special functions. 4. Compute the double and triple integrals. 5. Analyze scalar and vector fields, and apply concepts of gradient, divergence, and curl to interpret physical phenomena in vector calculus.		
Unit – I	First Order Ordinary differential Equations	6 Hours
Exact differential equations, Equations reducible to exact form. Linear differential Equations, equations reducible to linear form, Bernoulli's equation.		
Unit – II	Applications of Differential Equations	4 Hours
Applications of Differential Equations to Orthogonal Trajectories, Newton's Law of Cooling, Kirchhoff's Law of Electrical Circuits.		
Unit – III	Numerical solution of ordinary differential Equation	5 Hours
Picard's method, Euler's method, Modified Euler's method and Runge-Kutta method of fourth order.		
Unit – IV	Integral Calculus	5 Hours
Beta and Gamma functions, Differentiation under Integral Sign.		
Unit – V	Multiple Integrals	5 Hours
Double and Triple integrations, Double integration over a given region, Change of Order of integration, Change to polar coordinates.		
Unit – VI	Vector Calculus	5 Hours
Scalar and vector fields, gradient of scalar point function, directional derivatives, curl and divergence - physical interpretation, solenoidal and irrotational vector fields.		
Internal Continuous Assessment (ICA) ICA shall be based on student's performance during tutorial sessions and on completion of six assignments one on each unit.		
Text Books		
1. A Text Book of Applied Mathematics, P. N. and J. N. Wartikar, Vol.1, Pune Vidyarthi Griha Prakashan 2. Advanced Engineering Mathematics, H. K. Das, S. Chand Publications, Delhi 3. Engineering Mathematics (Volume), ITL Education, Cengage Learning 4. Engineering Mathematics, Ravish R Singh and Mukul Bhatt, McGraw-Hill		

Reference Books

1. Higher Engineering Mathematics (42nd Edition), B.S. Grewal Khanna Publications, Delhi
2. Engineering Mathematics, Srimanta Paland Subodh C. Bhunia, Oxford Higher Education
3. Higher Engineering Mathematics, Ramana B.V., Tata McGraw Hill New Delhi, 2010
4. Applied Mathematics-I, II, Kreyzig's, Wiley

e-Resources

- <https://www.youtube.com/playlist?list=PLM-jfaoaU5ixfcUcahxFeH7qVKY4oc3Ec>





WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR

(An Autonomous Institute)

First Year B. Tech. (Group C), Semester-II

CE2ES2 Basic Civil & Mechanical Engineering

Teaching Scheme		Examination Scheme	
Lectures	3 Hours/week	ESE	60 Marks
Practical	2 Hours/week	ISE	40 Marks
Credits	4	ICA	25 Marks

Introduction

The Basic Civil Engineering course offers an overview of key areas like infrastructure, surveying, construction materials, and foundation types. It introduces modern tools like GIS, GPS, and total stations, along with green building and environmental engineering concepts. The course equips students with essential skills for planning and managing civil projects.

The course on Basic Mechanical Engineering is aimed at providing him the knowledge in all those areas through demonstrations and shop instructions. This course is also aimed at providing the student the exposure to Engineering concepts and machines which will help him to adapt to the teaching which takes place in higher semesters.

Course Prerequisites

To effectively grasp the concepts in the Basic Civil and Mechanical Engineering course, students should have a foundational understanding of physics and general science from secondary education. Familiarity with basic mechanical concepts like force, pressure, energy, and simple machines will support learning about fluid machines, thermodynamic systems, and power transmission. A working knowledge of geometry, units, and measurement techniques is essential for understanding surveying principles and interpreting construction elements. Some exposure to everyday materials such as cement, bricks, metals, and tools will help in comprehending construction practices and machine operations. Basic arithmetic and algebra are important for performing calculations related to belt drives, energy equations, and field measurements.

Course Objectives

1. To introduce Civil Engineering sub-domains, roles of engineers, and basics of roads, railways, bridges, and transportation systems.
2. To develop skills in surveying, levelling, angular measurements, and use of GIS, GPS, and total station for accurate field data.
3. To provide knowledge of building elements, materials, foundation types, green building, and basic environmental engineering.
4. To provide students with a fundamental understanding of fluid machines including pumps, compressors, turbines, and thermodynamic principles
5. To introduce students to power transmission systems, basic machine tools, and metal joining processes used in mechanical engineering applications.



Course Outcomes		
At the end of this course, students will be able to: 1. Recognize Civil Engineering sub-domains, the role of Civil Engineers, and basic components of roads, railways, bridges, and transportation modes. 2. Describe principles of surveying, levelling, angular measurements, and apply GIS, GPS, and total station techniques in fieldwork. 3. Identify building components, materials, foundation types, green building concepts, and basics of environmental engineering. 4. Identify and explain the construction, working principles, and applications of various fluid machines and apply the first law of thermodynamics to simple systems. 5. Identify power transmission elements, perform basic calculations on belt drives, and explain the operations of machine tools and basic welding techniques.		
Section-I (Basic Civil Engineering)		
Unit – I	Introduction to Civil Infrastructure	6 Hours
Introduction to various sub domains of Civil Engineering. Role of Civil Engineer in various construction activities. Various modes of transportation. Introduction to roads: IRC classification, typical functional cross sections. Introduction to railways: cross section and components and their functions. Introduction to Bridges: cross section and components and their functions.		
Unit – II	Introduction of Surveying and Geospatial Technologies	6 Hours
General Principles of Surveying, Leveling and its types, Determination of reduced levels using auto level. Introduction to Geographic Information System (GIS), Fundamentals of Global Positioning System (GPS), Application of total station.		
Unit – III	Fundamentals of Building Construction and Environmental Engineering	9 Hours
Elements of a building substructure and superstructure (Foundation, Column, beam, plinth, lintel, chajja, roof, parapet, spout, etc.) and their functions. Building Materials: Use of basic and advanced materials: Cement, Concrete, bricks, aggregate, sand, stones, mortar, timber, reinforcing steel, smart and eco-friendly materials Types of foundation (Shallow and deep foundation) and their functions, Concept of Green building. Environmental Engineering: Water treatment systems; Solid waste management.		
Section-II (Basic Mechanical Engineering)		
Unit – IV	Fluid Machines and Fundamentals of Thermodynamics	10 Hours
Power absorbing devices: Pumps: definition, classification, construction and applications of reciprocating pump, centrifugal pump.		



Compressors: construction, working and applications of reciprocating compressor, rotary compressors (vane blower)

Power producing devices:

Turbines: construction, working and applications of Pelton-wheel, Francis and Kaplan turbines.

Fundamentals of Thermodynamics:

Definition of thermodynamics, thermodynamic Systems, surrounding, universe, types of systems, state of system, properties- intensive and extensive, thermodynamic equilibrium, process and cycle, Zeroth Law of thermodynamics. Work, heat, first law of thermodynamics, first law applied to flow processes, steady flow process, steady flow energy equation (SFEE) (Numerical on first law of thermodynamics)

Unit – V	Power Transmission Systems, Machine Tools, and Metal Joining Processes	10 Hours
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Belt drives: Open and cross belt drives, materials of belt, types of belts, length of belt for open and cross drive, velocity ratio of simple and compound belt drive, centrifugal tension, maximum power transmitted (numerical on simple belt drive only)

Other Transmission Systems: Chain drive, gear, types of gears (excluding gear terminology), gear trains- simple and compound, epicyclical gear train.

Machine tools:

Centre lathe: Basic elements, construction, working, and operations (turning and facing only).

Drilling machine: Basic elements, construction, working, and operations on pillar drilling machine (drilling and reaming only)

Metal Joining Processes:

Welding: Definition and types of welding processes: Manual Metal Arc Welding, Spot Welding

Internal Continuous Assessment (ICA)

The ICA shall consist of any 4(four) number of Practical's / Demonstrations for each section from the respective section list:

***Note:**

1. Practical's of Basic Civil Engineering and Basic Mechanical Engineering will be conducted in alternate weeks.
2. As a part of the completion of ICA, students shall submit separate completed Assignments, Journals and drawings at the end of the course.

List of Experiments

1. Study and Use of Auto Level for Determination of difference in Elevations.
2. Study and Use of Total station.
3. Determination of Latitude, Longitude, and Elevation Using GPS.
4. To determine the strength of coarse aggregates.
5. Exploring Building Materials: Visit and Lab Demonstration Experience.
6. Determination of Compressive Strength of Concrete Cube
7. Study and demonstration of vapour compression refrigeration cycle (e.g., domestic refrigerator and split AC).
8. Study and comparison of 2-stroke and 4-stroke engines
9. Study of pumps and compressors



10. Study of belt, chain, and gear drive mechanisms in power transmission systems.
11. Demonstration of turbines.

Text Books

1. Elements of Civil Engineering, S. S. Bhavikatti, New Age International Publishers
2. Building Construction and Drawing, Bindra and Arora, Dhanpat Rai Publications
3. Essentials of Civil Engineering, Rangwala, S. C, Charotar Publishing House
4. Basics of Remote Sensing & GIS, S. Kumar, University Sc. Press
5. Thermal Engineering, P.L. Ballaney, Khanna Publishers
6. Thermal Engineering, Domkundwar, Kothandaraman, Domkundwar, Dhanpat Rai & Co.
7. Elements of Workshop Technology, Vol-I and II, S.K. Hajra Choudhary, A K, Hajra Choudhary, Nirjhar Roy, Media Promoters & Publishers Pvt. Ltd.
8. Design of Machine Elements, V.B. Bhandari, Tata McGraw Hill Publications

Reference Books

1. Manual on Green Building, Kolhatkar
2. Energy-efficient buildings in India, Mili Majumdar, TERI Press
3. Construction Technology, Vol. IV, Chudley, R, Longman group, England Course Plan
4. Materials for Civil and Construction Engineering, Mamlouk, M. S., and Zaniewski, J. P., Pearson Publishers
5. Engineering Thermodynamics, P. K. Nag, Tata McGraw-Hill Companies
6. Mechanical Engineering Design, Joseph E. Shigley, Charles R. Mischke, Tata McGraw-Hill Company

e-Resources

- <https://youtube.com/playlist?list=PLM-jfaoaU5iyy18fs4C4JdCkkBNu8j8ss&si=m2uCzJSSvUOZRQtt>
- <https://youtube.com/playlist?list=PLM-jfaoaU5iztC05LVCaHzZCn1p0BpdNt&si=6CdqqGICWh-gkmNW>



**WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR****(An Autonomous Institute)****First Year B. Tech. (Group C), Semester-II****MA2ES3 Engineering Graphics and CAD****Teaching Scheme****Lectures** 2 Hours/week**Practical** 2 Hours/week**Credits** 3**Examination Scheme****ESE** --**ISE** 50 Marks**ICA** 50 Marks**Introduction**

Engineering Graphics and CAD has become an indispensable tool for engineers, technocrats, architects, surveyors, designers, and many other professionals in recent times. It is used to graphically convey the ideas and information necessary for the construction or analysis of machines, structures, and systems. This course focuses on creating orthographic and isometric drawings through a multidisciplinary approach.

Course Prerequisites

Students enrolling in Engineering Graphics and CAD should have basic familiarity with geometric shapes, spatial reasoning, and visualization skills, typically developed through mathematics and geometry studies at the higher secondary level. While no prior experience with technical drawing or CAD software is required, students should be comfortable interpreting 2D and 3D objects and have the patience and precision needed to represent them graphically. A keen eye for detail and interest in design-oriented thinking will help students effectively grasp orthographic and isometric projection principles and use computer tools for engineering drawing.

Course Objectives

1. To develop the ability to visualize objects in orthographic and isometric views.
2. To accurately draw orthographic and isometric views using standard conventions.

Course Outcomes

At the end of this course, students will be able to:

1. Draw orthographic views of objects.
2. Draw isometric views of objects.

Unit – I**Introduction to Methods of Projections****2 Hours**

Introduction to Methods of Projections.



Unit – II	Orthographic Projections	9 Hours
Developing orthographic views (front, top, and side) from given pictorial representations using the First Angle Projection method.		
Unit – III	Isometric Projections	9 Hours
Difference between isometric view and projection. Drawing isometric views from given orthographic projection. Introduction to computer-aided drafting (CAD) software.		
Internal Continuous Assessment (ICA) ICA shall be based on drawing orthographic and isometric views on drawing sheets.		
Text Books		
1. Engineering Drawing, D. N. Jolhe, TATA McGraw Publishing Co. Ltd. 2. Engineering drawing, N. D. Bhatt, Charotar Publishing House Pvt. Ltd. 3. A Textbook of Engineering Graphics, Natarajan K.V., Dhanalakshmi Publishers, Chennai 4. Engineering Graphics-I.M.L. Dabhade, Vision Publication		
Reference Books		
1. Engineering Drawing, N. S. Parthasarathy & VelaMurali, Oxford Publication 2. Engineering Drawing, K. L. Narayana & P. Kannaiah, Scitech Publication 3. Engineering Graphics, A. M. Chandra, New Age International Publishers		
e-Resources		
https://youtube.com/playlist?list=PLM-jfaoaU5ixN5WZtJTdzHJbJeApKOTjm&si=FEc1WgEO9Ljrd3I-		





WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR

(An Autonomous Institute)

First Year B. Tech. (Group C), Semester-II

GE2VS1 English for Employability Skills

Teaching Scheme		Examination Scheme	
Lectures	2 Hours/week	ESE	60 Marks
Tutorial	1 Hour/week	ISE	40 Marks
Credits	3	ICA	25 Marks

Introduction

In today's competitive and dynamic job market, technical knowledge alone is not sufficient to ensure employability. Employers increasingly seek graduates who possess excellent communication abilities, interpersonal skills, and emotional intelligence in addition to technical competencies. English for Employability Skills is designed to bridge this gap by enhancing students' soft skills, with a focus on communication, teamwork, leadership, time management, and personal development. The course empowers students to present themselves confidently, perform effectively in interviews and group discussions, and thrive in both academic and professional environments.

Course Prerequisites

To make the most of the English for Employability Skills course, students should have a basic proficiency in English language, including understanding simple instructions, forming coherent sentences, and reading comprehension. Prior exposure to general written communication—such as email writing or paragraph writing—is helpful. A foundational understanding of common social interactions and general etiquette aids in grasping interpersonal and soft skills. Additionally, basic computer literacy, including familiarity with presentation software like PowerPoint, is useful for developing presentation skills. Overall, a willingness to participate actively, self-reflect, and improve both spoken and written communication is essential as a prerequisite for this course.

Course Objectives

1. Develop students' proficiency in preparing and delivering effective presentations.
2. Equip students with the necessary skills to participate meaningfully in group discussions.
3. Enhance students' ability to handle personal interviews with confidence and competence.
4. Strengthen students' technical writing abilities, including resume, various applications and report writing.
5. Foster essential soft skills for personal growth and professional success.

Course Outcomes

At the end of this course, students will be able to:

1. Design and deliver professional presentations with clarity and impact.
2. Participate effectively and confidently in group discussions.
3. Perform successfully in various types of personal interviews.

4. Create impactful resumes and write technical reports, applications appropriately. 5. Demonstrate soft skills such as leadership, teamwork, time management, and emotional intelligence in real-life scenarios.		
Unit – I	Mastering Presentation Skills	6 Hours
Introduction Presentation, Importance of presentation Skills for Engineers, Structure of a Good Presentation, Use of AI for making PPT, 4 P's of Effective Presentation, Do's and Don'ts of Online and Hybrid effective presentation, Presenting PPT and Content delivery Skills		
Unit – II	Effective Group Discussion Skills	6 Hours
Introduction to Group Discussion, prerequisites of GD, Tips for successful GD, Do's and Don'ts of GD, Advanced strategies to overcome group discussion challenges		
Unit – III	Personal Interview skills	6 Hours
Fundamentals of Interviewing & Types, Preparation of PI, Effective Interview Techniques, Do's and Don'ts of personal Interview, Mock Interview & Feedback, strategic answering, FAQ's, Advanced strategies to overcome personal Interview challenges		
Unit – IV	Technical Writing Skills	6 Hours
ATS friendly Resume writing, Components of an Effective Resume, resume as per the Job Description, Writing various applications and Technical Reports writing		
Unit – V	Employability Skills	6 Hours
SMART Goal setting, strategies for effective time management, Team building and leadership, SWOC Analysis, Motivation and Positive thinking, Stress management and, Emotional intelligence.		
Internal Continuous Assessment (ICA) Students must complete at least 10 exercises from the list below during language lab sessions: 1. Prepare a PowerPoint presentation using AI and deliver on a selected topic. 2. Oral presentation on personal short-term and long-term goals using vision Board. 3. Participate in a group discussion and submit a written reflection on the topic. 4. Present and record the responses of frequently asked interview questions. 5. Prepare an ATS friendly resume tailored for campus placement opportunities. 6. Visit an industry and write a detailed technical report. 7. Conduct and document a personal SWOT analysis. 8. Prepare a case study on a company entering a global market and write about the risks it faced. 9. Analyse how teamwork contributed to the success of a challenging project. 10. Analyse the impact of different leadership styles on team motivation. 11. Participate in exercises focused on emotional intelligence and stress management. 12. Engage in creativity and entrepreneurship focused motivational tasks.		

Text Books

1. Soft Skills: An Integrated Approach to Maximize Personality, Gajendra Singh Chauhan & Sangeeta Sharma, Wiley India Pvt. Ltd.
2. Communication Skills for Professionals, Nira Konar, PHI Learning, 3rd Edition, 2022
3. On Writing Well, William Zinsser, Harper Resource Book, 2001
4. Technical English, Dr. M. Hemamalini, Wiley India Pvt. Ltd.
5. Professional Speaking Skills, Aruna Koneru, Paperback, January 2018
6. Group Discussion and Interview Skills, Priyadarshi Patnaik, Cambridge University Press India, 2nd Edition, 2015

Reference Books

1. Soft Skills, K. Alex, S. Chand Publications
2. Soft Skills: A Textbook for Undergraduates, Ajay R. Tengse, Orient Black Swan
3. Communication Skills, Sanjay Kumar & Pushpa Lata, Oxford University Press
4. Managing Soft Skills for Personality Development, B. N. Ghosh, McGraw Hill
5. Soft Skills for Everyone, Jeff Butterfield, Cengage Learning
6. Soft Skills for Managers, Dr. T. Kalyana Chakravarthi & Dr. T. Latha Chakravarthi, Biztantra Publication
7. Effective Communication, Adair, John, London: Pan Macmillan Ltd., 2003
8. Good English: Getting it Right, Ajmani J. C., Rupa Publications, New Delhi, 2012
9. Handling Tough Job Interviews, Amos, Julie-Ann, Jaico Publishing, Mumbai, 2004

e-Resources

- <https://youtube.com/playlist?list=PLM-jfaoaU5iyQgWy-WHgFc2G21J-agChw&si=kaCVsm5MJ-dSmtsT>





WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR
(An Autonomous Institute)
First Year B. Tech. (All Branches - Group B), Semester-II

CS2PC2L Python Programming

Teaching Scheme		Examination Scheme	
Lectures	2 Hours/week	ESE	50 Marks
Practical	2 Hours/week	ISE	25 Marks
Credits	3	ICA	25 Marks

Introduction

Python is a popular, general-purpose, open-source, multi-paradigm programming language known for its simple syntax, readability, and rich set of libraries. It supports rapid application development and is suitable for both interactive programming and large-scale software development. This course introduces the fundamentals of Python programming, including its powerful libraries and advanced features, enabling students to develop efficient solutions for computational problems.

Course Prerequisites

Basic knowledge of computer programming concepts

Course Objectives

1. To introduce Python programming constructs and data structures for problem solving.
2. To familiarize students with file handling, CSV processing, and exception handling techniques.
3. To impart object-oriented programming concepts for developing modular solutions.
4. To introduce students to Python modules, standard libraries, and advanced programming features.

Course Outcomes

At the end of this course, students will be able to:

1. Apply Python programming constructs and data structures to solve computational problems.
2. Implement file handling, CSV processing, and exception handling techniques in Python applications.
3. Apply Object-oriented features to provide solution.
4. Employ Python modules, standard libraries, and advanced programming features to develop efficient solutions for computational problems.

Unit – I

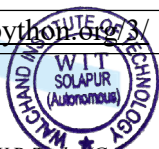
Introduction to Python

4 Hours

Features of Python, Installing Python, Python Interpreter and IDEs, Variables, Data Types, Input and Output Functions, Type Conversion, Operators and Expressions



Unit – II	Control Structures and Functions	6 Hours
Conditional Statements, Looping Constructs, Nested Loops, Break and Continue Statements, Functions, Arguments and Return Values, Scope of Variables, Recursive Functions		
Unit – III	Strings and Data Structures	6 Hours
String Operations and Methods, Lists, Tuples, Sets, Dictionaries, Basic Operations on Data Structures, List Comprehensions		
Unit – IV	Introduction to Object Oriented Programming in Python	6 Hours
Classes, Objects, Constructor & Self, Reference Variable, Abstraction & Encapsulation, Static variables & methods, Aggregation, Inheritance		
Unit – V	File Handling & Exception Management	4 Hours
Reading and Writing Files, File Modes, Working with Text Files, CSV File Reading and Writing Exceptions, Exception Handling using try-except-finally, User-defined Exceptions		
Unit – VI	Advanced Python Programming and Libraries	4 Hours
Modules and Packages, Introduction to Standard Libraries: Math, Datetime, Regular expression, Introduction to NumPy and Pandas, Lambda Functions, Map, Filter, Iterators and Generators, Collections Module (Counter, defaultdict, deque)		
Internal Continuous Assessment (ICA) ICA shall consist of a minimum of 10 practical assignments covering all the topics prescribed in the curriculum and one mini project demonstrating the application of Python programming concepts.		
Text Books		
1. Programming in Python 3, Mark Summerfield, Second Edition		
Reference Books		
1. Python Cookbook, David Beazley and Brian K. Jones, Third Edition, Shroff Publishers & Distributors Pvt. Ltd., ISBN :978-93-5110-140-6 2. Learning Python, Mark Lutz, 5th edition 3. Programming Python (English), Mark Lutz, 4th Edition 4. Testing Python, David Sale, Wiley India (P) Ltd., ISBN :978-81-265-5277-1		
e-Resources		
Python documentation - https://docs.python.org/3/		





WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR

(An Autonomous Institute)

First Year B. Tech. (Group C), Semester-II

GE2VS2L Workshop Practice

Teaching Scheme		Examination Scheme	
Lectures	1 Hour/week	ESE	--
Practical	2 Hours/week	ISE	--
Credits	2	ICA	50 Marks

Introduction

This course is designed to provide students with practical knowledge of computer hardware components, operating system installation, and productivity software. Emphasis is placed on hands-on experience, including identifying key PC components, setting up dual-boot systems, and working with Microsoft Office tools such as Word, PowerPoint, and Excel. The course aims to equip learners with essential digital skills required for academic, personal, and professional tasks in the modern technological environment.

Course Prerequisites

Students taking this course in the Workshop Practice course in Computer Science & Engineering or Information Technology should have a basic familiarity with computers and digital literacy skills, typically developed through school-level computer education. They should be comfortable navigating file systems, using a keyboard and mouse, and understanding the purpose of basic hardware components. No prior experience in operating system installation or office applications is required, but a logical approach, curiosity about how computers work, and a willingness to explore software tools are important. The course is best suited for students who are eager to build foundational digital competencies for academic and professional success.

Course Objectives

1. To make students learn the hardware and software of a computer.
2. Gain hands-on experience by installing and configuring operating systems, including dual-boot setups.
3. Develop skills in creating documents, presentations, spreadsheets, and other digital content using MS-Word, MS-PowerPoint, and MS-Excel, including advanced features and functions.

Course Outcomes

At the end of this course, students will be able to:

1. Identify various hardware and software components of a computer and compare between them.
2. Install and configure Operating systems, including dual-boot setups.
3. Design professional documents, presentations, and spreadsheets using MS-Word, MS-PowerPoint, and MS-Excel.



Unit – I	Identification of various PC hardware components	3 Hours
USB Mouse, PS/2 Mouse, Keyboard, LCD/LED Monitor, VGA, HDMI, CAT5, CAT6, server, routers, fiber cable, Hard disk, RAM, CMOS battery, SMPS, cache, ROM, BIOS		
Unit – II	Installation of Operating System	4 Hours
Install any two operating systems on a PC making it dual boot, including the latest version of Ubuntu Linux, Windows 10.		
Unit – III	Introduction to Microsoft Office: MS-Word, MS-PowerPoint	4 Hours
MS-Word: Overview, Creating a Simple Document, Formatting and Page Layouts, Printing Concept, Table of Content, Creating Greeting Card, Letterhead, Creating Flyer, Resume, Hyperlink and Bookmarks, Mail Merge. MS-PowerPoint: Overview, Inserting Slides, Text Typing and Formatting, Creating Digital Photo Album, Transition and Animation Effects, Automatic Timings and Action Buttons, Header and Footer, Designs and Themes		
Unit – IV	MS-Excel and Advanced Excel	3 Hours
MS – Excel: Overview of Excel, Formatting, Sum, Average, Min, Max, Len, Count, Upper, Lower, Page Layout Advanced Excel: Pivot Table and Chart, Planning Event Expenses, Creating Planner, Planning of Monthly Budget, Protect Sheet		
Internal Continuous Assessment (ICA) ICA shall be based on completion of any of the eight LAB Sessions based on following topics. 1. Identify various computer Hardware components. 2. Installation of Windows 10 Operating system and dual booting it. 3. Design a greeting card using MS-Word, incorporating images and text. 4. Create a flyer for an event, using MS-Word's formatting and design tools. 5. Use Mail Merge to create personalized letters to multiple recipients. 6. Design a presentation on a topic of your choice, using MS-PowerPoint's formatting and design tools. 7. Create a digital photo album using MS-PowerPoint, including transitions and animation effects. 8. Create a budget spreadsheet using MS-Excel, including formulas for calculations. (SUM, AVERAGE, MIN, MAX) 9. Create a dashboard using pivot tables and charts to analyze data. 10. Create a simple chart to visualize data.		

Text Books

1. Building Your Own Computer Made Easy: The Step By Step Guide (Computers Made Easy), James Bernstein
2. Raspberry Pi Cookbook: Software and Hardware Problems and Solutions, Simon Monk , 3rd Edition
3. Learn to use computer: A comprehensive guide for beginners, covering the basics of MS-Office Applications, MS-Word, PowerPoint and Excel for beginners, Inderjeet Singh





WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR

(An Autonomous Institute)

First Year B. Tech. (Group C), Semester-II

GE2MC1 Democracy, Elections and Good Governance

Teaching Scheme		Examination Scheme	
Lectures	--	ESE	50 Marks
Tutorial	--	ISE	--
Credits	--	ICA	--

Introduction

This course introduces students to the foundational principles and functioning of democracy in India, with a special focus on the role of elections and the framework of good governance. It explores the multidimensional nature of democracy, the constitutional basis for local self-governance, and citizen responsibilities in the democratic process. Additionally, it provides insights into governance reforms and initiatives aimed at enhancing transparency, accountability, and participatory administration.

Course Prerequisites

Before taking this course, students should possess a basic understanding of the Indian Constitution, including its fundamental rights and duties. Familiarity with the structure of the Indian government—such as the roles of the legislature, executive, and judiciary—is recommended. A general awareness of current national and local civic issues will also aid comprehension. Basic civic sense, awareness of voting rights, and an interest in public affairs or social issues will help students engage more meaningfully with topics like democratic participation, electoral processes, and governance reforms. An open and analytical mindset is beneficial for exploring the challenges and practical aspects of governance and democracy in India.

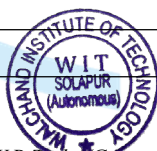
Course Objectives

1. To make the pupil aware of the importance of democracy.
2. To make the individual understand the different aspects of democracy and its implications in the overall development of the state.

Course Outcomes

At the end of this course, students will be able to:

1. Explain dimensions of Democracy and Challenges before Democracy
2. Explain duties of an individual towards electoral process
3. Express Concept and Good Governance initiatives in India



Unit – I	Democracy in India	
Dimensions of Democracy: Social, Economic and Political, Decentralization: Grassroots Level Democracy, Challenges before Democracy; women and marginalized sections of the society		
Unit – II	Election to Local Self Government Bodies	
73rd and 74th constitutional amendment Acts: Institutions at the local level and Role of State Election Commission, Local Body Elections, Duties of an Individual towards electoral process, Awareness of Voters		
Unit – III	Good Governance	
Meaning and concept, Government and Governance, Good Governance initiatives in India		
Text Books		
1. Who Wants Democracy? Alam Javeed, Orient Longman, Delhi, 2004 2. Why India votes?, Banerjee Mukulika, Routledge, Delhi, 2014 3. Good Governance: Never on India's Radar, Godbole Madhav, Rupa and Co., New Delhi, 2014 4. India After Gandhi: The History of the World's Largest Democracy, Guha Ramchandra, Pan Macmillan, New Delhi, India, 2017 5. Dr. Ambedkar and Democracy: An Anthology, Jaffrelot Christophe and Narendar Kumar (ed.), Oxford University Press, New Delhi, 2018 6. Local Governance in India: Decentralization and Beyond, Jayal Niraja, Amit Prakash, Pradeep Sharma, Oxford University Press, New Delhi, 2006 7. The Success of India's Democracy, Kohli Atul(ed.), Cambridge University Press, Cambridge, 2001 8. Indian Democracy, Oxford India Short Introductions, Palshikar Suhas, Oxford University Press, New Delhi, 2017 9. A Grammar of Democracy, Sawant P. B., Bhashya Prakashan, Mumbai, 2013 10. Indian Democracy, ICSSR Research Survey and Explorations Volume II, Suri K.C.(ed.), Oxford University Press, New Delhi, 2013 11. Indian Democracy: Meaning and Practices, Vora Rajendra and Suhas Plashikar(ed.), Sage Publications, New Delhi, 2001 12. हरबलेलेसुशासन, गोडबोलेसुजाता (अनुवादित), विश्वकर्मापब्लिकेशन्स, पुणे, २०१५ 13. भारतीयलोकशाही: अर्थआणिव्यवहार, लेलेचित्रा (अनुवादित), डायमंडपब्लिकेशन्स, पुणे, २०१० 14. गांधीनंतरचाभारत: जगातीलसर्वातमोठ्यालोकशाहीचाइतिहास, साठेशारदा (अनुवादित), मॅजेटिकपब्लिशिंगहाऊस, ठाणे, २०१७ 15. लोकशाहीजिंदाबाद, सोनावणेमनोहर(अनुवादित), समकालीनप्रकाशन, पुणे, २०१०		



**WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR****(An Autonomous Institute)****First Year B. Tech. (Group C), Semester-II****GE2MC2 Disaster Management**

Teaching Scheme		Examination Scheme	
Lectures	--	ESE	--
Practical	--	ISE	--
Credits	--	ICA	50 Marks

Introduction

Disaster Management is an interdisciplinary subject that deals with understanding disasters, their causes, impacts, prevention, preparedness, mitigation, response, and recovery. The course provides knowledge about natural and man-made disasters and emphasizes the importance of disaster risk reduction for sustainable development. It also introduces students to disaster management frameworks, policies, modern technologies, and the role of engineers, government agencies, and communities in minimizing disaster risks and ensuring effective disaster response and rehabilitation.

Course Objectives

1. To develop understanding of the concepts, types, causes, impacts, and management of natural and man-made disasters.
2. To familiarize students with disaster management cycle, risk assessment, mitigation, preparedness, response, recovery, and disaster management policies in India.
3. To explain the role of science, technology, engineering, and various agencies in disaster risk reduction, preparedness, and sustainable disaster management.

Course Outcomes

At the end of this course, students will be able to:

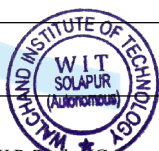
1. Understand and explain fundamental concepts of disaster management, types of disasters, their causes, impacts, and global trends.
2. Analyze and apply disaster management cycle principles including risk assessment, mitigation, preparedness, response, and recovery strategies.
3. Evaluate and develop disaster management strategies using policies, institutional frameworks, and modern technologies for real-world applications.

Unit – I**Understanding Disasters****2 Hours**

Understanding the Concepts and Definitions of Disaster, Hazard, Vulnerability, Risk, Capacity – Disaster and Development, and Disaster Management



Unit – II	Types, Trends, Causes, Consequences and Control of Disasters	8 Hours
Geological Disasters (earthquakes, landslides, tsunamis, mining); Hydro-Meteorological Disasters (floods, cyclones, lightning, thunder-storms, hail storms, avalanches, droughts, cold and heat waves) Biological Disasters (epidemics, pest attacks, forest fire); Technological Disasters (chemical, industrial, radiological, nuclear) and Man-made Disasters (building collapse, rural and urban fire, road and rail accidents, nuclear, radiological, chemical and biological disasters) Global Disaster Trends – Emerging Risks of Disasters – Climate Change and Urban Disasters		
Unit – III	Disaster Management Cycle and Framework	8 Hours
Disaster Management Cycle – Paradigm Shift in Disaster Management, Pre-Disaster – Risk Assessment and Analysis, Risk Mapping, Zonation and Microzonation, Prevention and Mitigation of Disasters, Early Warning System; preparedness, Capacity Development; Awareness During Disaster – Evacuation – Disaster Communication – Search and Rescue – Emergency Operation Centre – Incident Command System – Relief and Rehabilitation Post-disaster – Damage and Needs Assessment, Restoration of Critical Infrastructure – Early Recovery – Reconstruction and Redevelopment IDNDR, Yokohama Strategy, Hyogo Framework of Action		
Unit – IV	Disaster Management in India	7 Hours
Disaster Profile of India – Mega Disasters of India and Lessons Learnt Disaster Management Act 2005 – Institutional and Financial Mechanism National Policy on Disaster Management, National Guidelines and Plans on Disaster Management Role of Government (local, state and national), Non-Government and Inter-Governmental Agencies		
Unit – V	Applications of Science and Technology for Disaster Management	7 Hours
Geo-informatics in Disaster Management (RS, GIS, GPS and RS) Disaster Communication System (Early Warning and Its Dissemination) Land Use Planning and Development Regulations Disaster Safe Designs and Constructions Structural and Non-Structural Mitigation of Disasters S&T Institutions for Disaster Management in India		
Unit – VI	Suggested Areas for Project and Assignments	2 Hours
Study of Recent Disasters (at local, state and national level) and Preparation of Disaster Risk Management Plan of an Area or Sector Role of Engineers in Disaster Management		



Possible Reading Material:

1. Coppola D. P., 2007. Introduction to International Disaster Management, Elsevier Science (B/H), London.
2. Manual on Natural Disaster Management in India, M. C. Gupta, NIDM, New Delhi.
3. An Overview on Natural & Man-made Disasters and Their Reduction, R. K. Bhandani, CSIR, New Delhi.
4. World Disasters Report, 2009, International Federation of Red Cross and Red Crescent, Switzerland.
5. Encyclopedia of Disaster Management, Vol. I, II and III – Disaster Management Policy and Administration, S. L. Goyal, Deep & Deep Publications, New Delhi, 2006.
6. Encyclopedia of Disasters – Environmental Catastrophes and Human Tragedies, Vol. I & II, Angus M. Gunn, Greenwood Press, 2008.
7. Disasters in India: Studies of Grim Reality, Anu Kapur and others, 2005, Rawat Publishers, Jaipur.
8. Management of Natural Disasters in Developing Countries, H. N. Srivastava and G. D. Gupta, Daya Publishers, Delhi, 2006.
9. Natural Disasters, David Alexander, Kluwer Academic, London, 1999.
10. Disaster Management Act 2005, published by Government of India.
11. Publications of National Disaster Management Authority (NDMA) on various templates and guidelines for Disaster Management.
12. NIDM Publications.
13. High Power Committee Report, 2001, J. C. Pant.
14. Disaster Mitigation in Asia & Pacific, Asian Development Bank.
15. National Disaster Management Policy, 2009, Government of India.
16. Disaster Preparedness Kit, American Red Cross.

